

GenAI Evaluation Metrics

A simple validation guide for LLM applications such as chatbots, summarization, classification, report generation, email drafting, and enterprise productivity use cases.

Purpose: This PDF explains the most important GenAI evaluation metrics in a practical order, so participants can understand how to validate whether a GenAI application is accurate, useful, safe, compliant, and ready for real enterprise usage.

1. Most Important GenAI Metrics to Remember

For most GenAI applications, start with these core metrics. They are easy to explain and useful across banking, HR, operations, analytics, customer support, and knowledge-work use cases.

Metric	Simple Meaning	Why It Matters
Accuracy	Is the output correct?	Prevents wrong business decisions.
Relevance	Does the output answer the user request?	Avoids off-topic or generic answers.
Completeness	Does the output cover all required points?	Prevents missing important information.
Factual Correctness	Are the facts true and verifiable?	Reduces false or misleading content.
Instruction Following	Did the model follow the prompt rules?	Important for enterprise workflows and automation.
Format Compliance	Is the output in the requested format?	Needed for JSON, tables, reports, emails, and APIs.
Safety & Compliance	Is the output safe, private, and policy-compliant?	Critical for regulated domains such as banking.
Hallucination Rate	Did the model invent unsupported information?	Measures one of the biggest GenAI risks.

2. GenAI Metrics with Simple Questions

Use these questions during validation, demo review, UAT, or production monitoring.

Metric	Simple Validation Question
Accuracy	Is the answer or output correct?
Relevance	Does it directly answer the user question?
Completeness	Did it include all required information?
Coherence	Is the answer logically structured and easy to follow?
Fluency	Is the language clear, professional, and readable?
Factual Correctness	Can the facts be verified from a trusted source or reference answer?
Hallucination Rate	Did the model add anything that was not true or not supported?
Instruction Following	Did it follow all rules given in the prompt?
Format Compliance	Did it return the required structure, such as JSON, table, or email?
Consistency	Does it give stable answers for similar inputs?
Bias/Fairness	Does it avoid unfair or discriminatory outputs?
Privacy	Does it avoid exposing personal or confidential information?
Toxicity/Safety	Does it avoid harmful, abusive, or unsafe content?
Latency	Does it respond within acceptable time?
Cost	Is the token/API cost acceptable for business usage?

3. Simple GenAI Evaluation Flow

A good GenAI validation process should check quality, factuality, safety, and operational readiness.

Step	What to Validate	Example Checks
Step 1: Define Use Case	Clarify what the GenAI app should do.	Summarization, classification, report writing, Q&A;, email drafting.
Step 2: Prepare Test Data	Create realistic input examples.	Normal cases, edge cases, confusing inputs, unsafe inputs, missing-data cases.
Step 3: Evaluate Output Quality	Check usefulness and readability.	Accuracy, relevance, completeness, coherence, fluency.
Step 4: Evaluate Factuality	Check whether the output is true.	Factual correctness, hallucination rate, citation/source validation.
Step 5: Evaluate Prompt Compliance	Check whether instructions are followed.	Format compliance, tone, length, JSON schema, required sections.
Step 6: Evaluate Safety	Check business and policy risks.	Privacy, toxicity, bias, compliance, confidential data exposure.
Step 7: Evaluate Operations	Check production readiness.	Latency, cost, failure rate, escalation rate, user feedback.

Simple rule: Do not validate only the final answer. Validate whether the output is useful, true, safe, consistent, and suitable for the business workflow.

4. Explanation of GenAI Evaluation Metrics

Accuracy

Accuracy checks whether the GenAI output is correct for the given task. For classification, it means the predicted class is correct. For Q&A, it means the answer matches the expected answer. Example: If a complaint should be classified as High Priority and the model predicts High Priority, it is accurate.

Relevance

Relevance checks whether the output is related to the user request. Sometimes an answer may be grammatically correct but still not answer the actual question. Example: If the user asks for key risks in a banking report, the answer should focus on risks, not general banking definitions.

Completeness

Completeness checks whether all required points are included. Example: A good customer complaint summary may need issue, customer impact, root cause, action taken, pending owner, and SLA status. Missing any major point reduces completeness.

Coherence

Coherence checks whether the answer is logically organized. The answer should have a clear flow and should not contradict itself. This is important for reports, executive summaries, and customer-facing responses.

Fluency

Fluency checks language quality. The output should be readable, professional, and grammatically correct. For enterprise use, fluency matters because poor language can reduce trust.

Factual Correctness

Factual correctness checks whether the information in the output is true. It is usually validated against source documents, a reference answer, expert review, or trusted data. This is critical when the output influences decisions.

Hallucination Rate

Hallucination rate measures how often the model invents information. Example: If a policy does not mention a refund, but the model says the customer will receive a refund, that is hallucination.

Instruction Following

Instruction following checks whether the model followed the prompt rules. Example: If the prompt says "answer in 5 bullet points" or "do not mention customer name", the model should follow those instructions exactly.

Format Compliance

Format compliance checks whether the output follows the requested structure. This is important when GenAI output is used by another system. Example: valid JSON, a fixed table format, a standard email template, or a required report structure.

Consistency

Consistency checks whether the model gives stable outputs for similar inputs. A production application should not give very different decisions for nearly identical cases unless there is a valid reason.

Bias and Fairness

Bias and fairness checks whether the model treats users, customers, or groups unfairly. Example: A loan-support assistant should not make biased assumptions based on gender, region, language, or background.

Privacy and Data Leakage

Privacy checks whether the model avoids exposing personal, confidential, or sensitive information. Example: It should not reveal account numbers, passwords, private customer details, or internal-only content.

Toxicity and Safety

Toxicity and safety checks whether the output avoids harmful, abusive, unsafe, or inappropriate content. This is important for public-facing chatbots and internal enterprise assistants.

Latency

Latency measures how long the GenAI system takes to respond. Even a high-quality answer may not be acceptable if it is too slow for customer support or operational workflows.

Cost per Request

Cost per request measures token cost, API cost, infrastructure cost, and tool usage cost. This helps decide whether the GenAI application can scale economically.

5. Task-Specific GenAI Metrics

Different GenAI use cases need additional task-specific metrics.

Use Case	Important Metrics	Example
Summarization	Coverage, conciseness, factual correctness, no hallucination	Does the summary include all important points without adding false information?
Classification	Accuracy, precision, recall, F1 score, confusion matrix	Did it correctly classify complaints by risk level or category?
Sentiment Analysis	Accuracy, precision, recall, F1 score by class	Did it correctly identify positive, neutral, or negative customer sentiment?
Chatbot	Task completion rate, resolution rate, escalation accuracy, user satisfaction	Did the chatbot solve the user issue or escalate correctly?
Report Generation	Completeness, structure, factual correctness, business relevance	Did the report include findings, risks, actions, and recommendations?
Email Drafting	Tone, clarity, instruction following, privacy, professionalism	Is the email polite, clear, and safe to send?
Code Generation	Correctness, test pass rate, security, maintainability	Does the generated code run and pass test cases?

6. Practical Banking Example

Use case: A GenAI assistant summarizes customer complaints for a banking operations manager.

Metric	Validation Question
Accuracy	Did it correctly identify the complaint issue?
Completeness	Did it include customer impact, product, issue type, SLA status, and pending action?
Factual Correctness	Did it avoid adding details not present in the complaint text?
Instruction Following	Did it follow the required management-summary format?
Privacy	Did it avoid exposing unnecessary customer PII?
Tone	Is the language professional and suitable for management?
Escalation Safety	Did it flag high-risk or urgent issues correctly?

7. Recommended Validation Set

For a practical GenAI application, prepare a test set with different input types: easy cases, complex cases, ambiguous cases, missing-information cases, unsafe requests, privacy-risk cases, and edge cases. Then score each output using the selected metrics.

Minimum enterprise validation: Accuracy + relevance + completeness + factual correctness + instruction following + privacy + safety + latency + cost.